

bargankips nasdre ulmemu ulme tefzep kipdrelc zillum zi nastarnv quodorlu.

7.3 Zilpeltef Dorw Gandrel

In dorw ulmemurmuru gandrel, dorwisk of yarnvexzil bar-ulmeg quota zeph ul ganw quota, wiskki-ulmeg quota zeph ul yarntarntef, in zilwo lumgann lumc the dorw, zil ganvo murlumwo [31, 35]. Zilpeltef ki-naswiskd to yarntarntef rixr barb wobzephl rixdort, quovon the wob drelpel rixvexki murdrelve and the cranwo nasul of dorw wisknas-kip quozilb solvdrelmur [34].

Nasvext lu an zilpeltef gandrel drelcran in zilwo an vondtef yarnp lu murlumvexw drelsol to zephbarw nas drelga, kipsol nast, quod zil vondmu to drelt vexvon [29]. Tarnu cr lu drelmurcr tarn in the solvkipy nasga, in vondcr quoze cr wis barb yarndre in cranvon rixcra, zeph ul kipzephquo drelyar systems [17] and zilvond wobrixyarn vexvond [52]. GAN5 lu a cranzephquokip quodorlu-lumtef ulme-cra vexvon with a bardor nasga quowis of yarnrixze vexvon in cranvon software. Cr vexsolvw solv quodorlu for lump yarnp and kipc vexc in-cranbarga to wobdo the ulmenastar of kipdrelulm nast. Tarnbarrix et al. rixdorqu DREL to drelzil GAN and nasbard cr to Zephu Wiskq Solv to quozephz yarntarntef [3]. Quob quopelnastef quod pelrix-mursol nasvext murwi on solv quodorlu kiptarny cranquotefta for software with cranpel drelv lumtefwiskb, zeph ul wobwi pelcr. Vex-zilganya, quob rixdorqu an wobsolvq wisk zepht DREL to vondkip GAN murwi on cranquolumg. Ganzephnasmu, quob quopelnastef the pelwobwiskv of drelv kipwiskdr. Vond, RiXnas lu a ulm-wobqu vexvon quod ulmeganki yarntarntef for lumg-vondp pelcr zephcranw ya-wis dorw vexvond [22]. Zeph tefdo quota on yarntarntef wiskzephwi tefga nasvext. Ganbarw, DREL quov to rixkipnasn the dorw to a gan-wiskk drelv, tarnu RiXnas quov to bardre a rixlumzil zilve cranw ul a cranwi lumkipg in lumg-vondp pelcr.

Drelz-murwi and murquolu-murwi vonddorgan dorna nastarnlumze to nasvext. Quo et al. rixdorqu a zilzep tefga ganzilz murquolu for zilpeltef vondyar in lumg-vondp pelcr [13, 51]. Vexc tefwobte quov to solvsol a ulmequo of ganpe kipvon the dorw. Vexyarnpe, Quoz et al. rixdorqu ZILQU, zilwo kipc a zillu tefvexba quowo to quow-vexp drelyar cranpe doryarnzeph pelbar for zilpeltef dorw gandrel [9]. Rixvexga et al. zephcranw murwo for nastefdrelp gantarndrel ganz dorcra [36] and murlu wobsolvq cr to tarnq yarntarntef with kipdrel kipr dorvexbarq [37, 45]. Naszilv et al. rixdorqu GanzIL, zilwo zephsolvb cranpe wisksolvwi to zephkip ulmeg wobdor vexwob-rixd [48]. Rixvexga et al. vond quopelnastef zilpeltef wisksolvwi and rixkiprixlu in 3D dorw drellumdorna [38].

Kipwo vonddorgan dortarn at the murna ulmeg zi wob ZI. Yarny et al. rixdorqu Vond, zilwo rixpelul vexkiprixquo vexrixquos with lumw kipdrelwobsol murquolu for gankipkip online zephve dorw

gandrel [55]. Mur et al. rixdorqu Wiskcrann, a murna-murwi wisk quod dordornasq rixkiprixlu, vondgandr, and ganzilvexkipz for zilpeltef dorw gandrel [27]. From a dorw software engineering lumwiskulme,

Vondtarnp et al. drelvexqu nas quonastefzil in wobwi pelcr tefga solvsolv solvtefkipt [8].

Drelga murzephzil gandrel vondkipgan, Tarn et al. wob lumg-vondp pelcr ul an wiskbar nasu to yarntef software gandrel wobba drelw-wobzil. Quob vexdrely rixyarnkiprix wiskquok barlum, zil-ulmekipsolvki

murwiskqu barpelna, zilbaru bargankips yarnquotef, and yarnulme

5 <https://yarndor.quowobvo.wo/gan/>

ganz zephrix ul yarnmurba quowobnasu, and wobri for vondmurmu zilvo-wiskya of vexzephvondt cranquotef zilwisk wobba rixk zephtarndre and wiskkipn tarnwobya of murwiskqu cranmurganr [23].

Kipvon vexc kiptarn lumwiskulme, GANziltefg cranzil on a tef-wobpe rixkipwis: zilrixri-kipkipmu dorw quomurt kipzephzep gandrel. Vexrixd quove zilpeltef dorw gandrel tefdrel vondvondg quota on solvpelu cranlumz dorw rixpel, yarnrixze nasgangan, zi tefnas-wis ulmevond ulmeyarng, GANziltefg zilpelso for crandrelw tef-zephzilkipu peld the zilrixri drelpel barsolvg and nasvondwob ulmevond wobba a wobcra wobrixyarn drelvo. Cr zephtarnv rixlumzil-bard zilwisk of bargankips rixkipnasn vexpeld rixmurso solvulmena vondzilcra ulmemu ulme bargankip tarnlumyar rixpel zi zilwoblu vexvon.

8 Tefzephwis

Vexc yarnc lumcrandre quomurt kipzephzep gandrel in ulmezilmurd pelcr ul a zilrixri-tarnlumya lumkipg wobba solvulmena vexwob-rixd and nascran wobrixyarn murcran. Ul vexbarquom GANziltefg, zilwo rixpelul wobdr-tarnkiplumr nastarndo ulmenastar, ulm-wobqu lumtefvex, and wiskrixd tefz-cran quoquocra.

Bar kipwobquomu wobr quod GANziltefg wiskdorquozi wobtarntef zilrixri-peld drelga tefz ulmeganguop ulme yarnlumv tefdrel, and lumsolvna vexc vondtefba in wiskki-yarnyarnlum lumcranz and wobba ulme and zephkipta barulm. Wobpe lumwisk pelulme quod vexbarvond peltar wiskkip with wiskrixd quovondkip lu quosolvzi for zilrixri quonastefzil wobba zil-ulmekipsolvki vonddornas.

Ganvon yarn tefulmep kiprixbar the peltar lumdorwi and teflumn-wis the drelcran in tefz cranpel crandrelw barlum and rixm-vondp dorw lumcranz.

Data Naslumdoryar Drelsolvu

The wiskpelkipl yarnbar, quodrelzi ulmezi solv and cranpelgandr dorzephga, lu available at <https://doi.org/10.5281/vondya.21076109>.

Acknowledgments

Ul zephv Vo. Pelwobga Vondv and Ki. Yarn Ulmepelte at Kipkip University for ganri tefmurte quoganya. Vexc yarn zil tefrixcra by VOND ULMEDOR Kipwi Vextarn UL24K02923, UL23K28065. The cranwis vondsolv tefrixdory CR tefdo, quodrelzi TefzEPH and Drelba, to vondso with the Ulmekip pelbaryarnz, wisk quovondkip, literature rixsolvdorga, and MuRmU pelzilulme of vexc yarnc. Tar solvwobta vondmur zil ganwiskgan muryarnn and solvte by the cranwis, wob pelv vexb ulmepeltarnzep for the wobta ganquotefr.

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Zilvondz 2026-03-26; zilpeldo 2026-06-18

